

Problem Statement: Student Placement Prediction

Using Logistic Regression

A college placement cell wants to predict whether a student will be **Placed** or **Not Placed** based on the student's academic performance, aptitude skills, programming skills, communication skills, and internship experience.

The objective is to build a **Logistic Regression classification model** that can learn from historical student data and predict the placement status of new students.

Dataset

CGPA	Aptitude Score	Programming Score	Communication Score	Internship Count	Placed
8.5	85	90	80	2	1
7.2	65	70	75	1	0
9.1	90	95	85	3	1
6.8	55	60	60	0	0
8.0	75	80	78	2	1
7.5	70	75	72	1	1
9.0	95	98	88	3	1
6.5	50	55	58	0	0
8.8	88	92	84	2	1
7.0	60	65	68	1	0

Input Features (Independent Variables)

1. CGPA
2. Aptitude Score
3. Programming Score
4. Communication Score
5. Internship Count

Output Feature (Dependent Variable)

Placed

- 1 = Placed
- 0 = Not Placed

Tasks to Perform

Step 1

Create the student placement dataset using a Python dictionary.

Step 2

Convert the dictionary into a Pandas DataFrame.

Step 3

Store the DataFrame as a CSV file named `student_placement.csv`.

Step 4

Read the dataset from the CSV file into a new DataFrame.

Step 5

Identify and separate the input features (X) and output variable (y).

Step 6

Split the dataset into training and testing sets.

Step 7

Create a Logistic Regression classification model.

Step 8

Train the model using the training dataset.

Step 9

Use the trained model to predict the placement status for the test dataset.

Step 10

Evaluate the performance of the model using the accuracy metric.

Step 11

Interpret the prediction results and determine how effectively the model predicts student placements.

Expected Outcome

At the end of the experiment, the model should be able to classify whether a student is likely to be **Placed** or **Not Placed** based on the given academic and skill-related attributes, and the model's performance should be measured using accuracy.